1. Hot water at 98°C flows through a steel pipe (thermal conductvity 54 W/mK), with ID 5.25 cm, OD 6.03 cm. The inside heat transfer coefficient is 1960 W/m^2K. The tube is exposed to atmospheric air at 20 °C , the outside heat transfer coefficient is 8 W/m^2K. Calculate the overall heat transfer coefficient based on the inside surface area and outside surface area. Consider an inside fouling factor of 0.0002 m^2K/W and an outside fouling factor of 0.0005 m^2K/W.

2. A heat exchanger is used to heat water at a rate of 0.8 kg/s from 30 °C to 80 °C, with hot oil entering 120 °C and leaving at 85 °C. The overall heat transfer coefficient based on the outside surface area shall be assumed as 125 W/m^2K. Calculate the heat exchanger area required considering (a) The heat exchanger is of parallel flow arranged (b) The heat exchanger is of counter flow arranged (c) The heat exchanger is of shell and tube arranged, with two shell pass and four tube passes. (water on shell side).

3. An oil cooler for a large diesel engine is to cool engine oil from 60 °C to 45 °C using sea water at an inlet temperature of 20 °C with a temperature rise of 15 °C. The design heat load is 140 kW. The overall heat transfer coefficient based on the outside diameter shall be assumed as 70 W/m^2K. Calculate the heat transfer surface area for single pass counter flow and parallel flow arrangements.

4. A counter flow heat exchanger of heat transfer area $12.5m^2$ is to cool oil (c_p= 2000 J/kgK) with water (c_p= 4170 J/kgK). The oil enters at 100°C and flow rate 2 kg/s, while water enters at 20°C and 0.48 kg/s. The overall heat transfer coefficient is 400 W/m^2K. Calculate the exit temperatures and heat transfer rate. (Ans. 131 kw, 85.6C)

5. A two shell pass four tube pass heat exchanger with area 26m^2 is to cool oil at 1.5 kg/s,(c_p= 2000 J/kgK) with water (c_p= 4170 J/kgK). The oil enters at 90°C and , while water enters at 19°C and 1.0 kg/s. The overall heat transfer coefficient is 250 W/m^{2}K. Calculate the exit temperatures and heat transfer rate. If the Oil flow rate is reduced to 75% of the initial flow rate, predict the exit temperatures with the same water flow rate. (Thout-40°C)

6. Water at the rate of 68 kg/min is heated from 35 to 75°C by an oil having a specific heat of 1.9 kJ/kg°C. The fluids are used in a counterflow arrangement. Oil enters the exchanger at 110°C and leaves at 75 °C. The overall coefficient is 320 W/m^{2}K. Calculate the heat exchanger area. (LMTD 37.44 °C, A 15.82 m$^{2.}$)

7. If the heat exchanger in the above case is a single shell two tube pass heat exchanger. (Water -shell side, Oil – tube side). Assume overall coefficient as 320 W/m^{2}K. Calculate the heat exchanger area (19.53 m^2)

8. Water at the rate 3.783 kg/s is heated from 37.78 to 54.44°C, in a shell and tube heat exchanger. On the shell side one pass is used with water as the heating fluid (1.892 kg/s), entering at 93.33 °C. The overall heat transfer coefficient is 1419 W/m^2°C, and the average velocity inside the tube is limited as 0.366 m/s. Also the tube length is to be limited as 2.438m. Calculate the number of tube passes, number of tubes per pass length of tube designed. (Number of tubes per pass 36, number of pass 2, length of tube 1.646 m)

9. A cross flow heat exchanger is to be designed to heat oil (c=1.9 kJ/kg°C) from 15 °C to 85 °C. Steam is used to heat oil, entering at 130 °C and leaving at 110 °C with flow rate of 5.2 kg/s. The overall heat transfer

coefficient is 275 W/m^2°C, and c -steam 1.86 kJ/kg°C. Calculate the area of heat exchanger. Assume Steam mixed and Oil unmixed. (A-10.82 m2)

10.In the problem 6, if only 40 kg/min of water is used with the same oil flow rate, predict the water outlet temperature for the same heat exchanger. Also calculate the heat exchange. (Water exit, 90.8 °C, Q- 155.5 kW).

11. Hot oil at 100°C is used to heat air in a shell and tube heat exchanger. The oil makes six passes and air makes one shell pass. 2.0 kg/s of air is to be heated from 20 to 80 °C. Oil specific heat 2100 J/kg°C and flow rate is 3 kg/s. Calculate area required for the heat exchanger for U= 200 W/m^2°C. (A-20.09 m2)

12. A shell and tube heat exchanger is used as an ammonia condenser with ammonia vapor entering shell at 50 °C as saturated vapor. Water enters the single pass tube arrangement at 20 °C and total heat transfer required is 200 kW. Overall heat transfer coefficient is 1000 W/m^2°C. Determine area required to achieve an effectiveness of 60%. with an exit water temperature of 40 °C. What percent reduction in heat transfer would result if the water flow rate reduced by half., keeping area and U same ?. (A=9.16m2, q new- 126 kW, % reduction -37%).